# Data Types

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# Data Type and Operations

[Adhikari 2017], [Hunt 2019]



- Every value has a type
  - The built-in type function returns the type of the result of any expression
    - *type(3)*
    - *type("Hello")*
- The type of an item of data (such as a string) determines what it is legal to do with the data and what the effect of various actions will be
  - E.g., the effect of using a **+** operator depends on the types involved
    - It adds numbers but concatenates strings
      - *3 + 4* *# Addition*
      - *"Hello " + "World"* *# Concatenates*

# Dynamic Typing

[Hunt 2019]



- Python has dynamic typing
  - The type of the data held by a variable can dynamically change as the program executes
    - *x = 1*
    - *x = "Hello"*
- It is not the approach used by many statically typed programming languages
  - e.g., *Java* and *C#*
- Both approaches have their pros and cons
  - The flexibility of Pythons variables are one of its major advantages

# Data Types in Python



- Numbers
  - Int
  - Float
- String
- Boolean
- None
- Collection Types
  - Tuple
  - List
  - Set
  - Dictionary

# Numbers

[Adhikari 2017]



- Computers are designed to perform numerical calculations
- Python distinguishes between two different types of numbers
  - Integers are called int values
    - *3, -5, 0*
  - Real numbers are called float values
    - *3.5, -4.333, 0.0*
- Converting to int or float
  - *int("3")*
  - *float("3.4")*

# String

[Adhikari 2017]



- Much of the world's data is text
- A piece of text represented in a computer is called a string
  - A string is a series, or sequence, of characters in order
  - Single and double quotes can both be used to create strings
    - *'hi'* and *"hi"* are identical expressions
    - It is useful if your string needs to contain one of the other type of string delimiters
  - Triple quotes allow multiline strings
    - *s = """This is a multiline string."""*
  - We can also define an empty string
    - *s = ""*

# Some String Operations

[Hunt 2019]



- String concatenation
  - *s = “Hello ” + “World”*
- Getting the length of a string
  - *len(s)*
- Accessing a character
  - Strings are indexed from zero
  - *s[3]*
  - *s[-2]*
- Accessing a subset of characters
  - *s[2:5]*
  - *s[:5]*
  - *s[5:]*
- Repeating strings
  - *“Hi” * 10*
- Converting to string
  - *str(3)*

# String Methods

[Adhikari 2017]



- Methods are functions that operate on objects
  - We will learn more about methods later
- These methods are called by placing a dot after the (string) object, then calling the function
  - *"loud".upper()*
  - *“Hello”.replace(“e”, “a”)*

# Some String Methods

[Hunt 2019]



- Splitting strings
  - *s.split(“ “)*
- Counting in strings
  - *s.count(“a”)*
- Replacing strings
  - *s.replace(“x”, “y”)*
- Finding sub strings
  - *s.find(“hi”)*
- Formatting string
  - *“The name is {}”.format(“John”)*
  - *“The name is {‘John’}.”*

# Mutable and Immutable Types

[Reitz 2016]



- Python has two kinds of built-in or user-defined types
- Mutable types allow in-place modification of the object's content
  - Such as lists
    - *my_list = [1, 2, 3]*
    - *my_list[0] = 4  # The new list is [4, 2, 3]*
  - They have mutating methods like *list.append()* that can modify the object in place
- Immutable types do not allow in-place modification of the object's content
  - Such as strings
    - *my_string = "123"*
    - *my_string[0] = "4"  # It generates an error*
  - These types provide no method for changing their content in place
    - We have to create another variable to store the modified object

# Immutability

[Hunt 2019]



- Strings are immutable (not changable)
  - Once a string object has been created, it cannot be changed in place
- If you try to change a string, you will create a new string
  - You will not affect the original string in anyway
  - Remember to store the result
    - *s = s.replace(“x”, “y”)*

# Boolean

[Hunt 2019], [Adhikari 2017]



- A Boolean type can only be one of ***true*** or ***false***
  - The Boolean type is actually a sub type of integer
    - It is easy to translate between these two types using *int()* and *bool()* functions
  - Python is quite in representing ***true*** and ***false***
    - ***0***, ***“”*** (empty strings), ***None*** equate to ***false***
    - Non-zero, non-empty strings, any object equate to ***true***
  - It is recommend to just use ***true*** and ***false***
    - It is often a cleaner and safer approach
- Boolean values most often arise from comparison operators
  - The comparison *3 > 2* generates ***true***

# None

[Hunt 2019]



- Python has a special type, the NoneType, with a single value, ***None***
  - This is used to represent null values or nothingness
  - It is not the same as False, an empty string, or *0*
    - it is a non-value
- It can be used to initiate a variable that we do not have an initial value for
  - *winner = None*
  - You can then test for the presence of *None* using *"is"* and *"is not"* operators
    - *print(winner **is** None)*

# Identity Operators

https://www.w3schools.com/python/python_operators.asp

https://realpython.com/python-is-identity-vs-equality/



- Identity operators are used to compare the objects
  - Not if they are equal, but if they are actually the same object, with the same memory location

| Operator | Description | Example |
|---|---|---|
| *is* | Returns ***true*** if both variables are the same object | *x is y* |
| *is not* | Returns ***true*** if both variables are not the same object | *x is not y* |

# Collection Types

[Hunt 2019]



- A collection is a single object representing a group of objects
  - It is also referred to as containers
- These collection classes are often used as the basis
  - For more complex or application specific data structures and data types

# Collection Types in Python

[Hunt 2019]



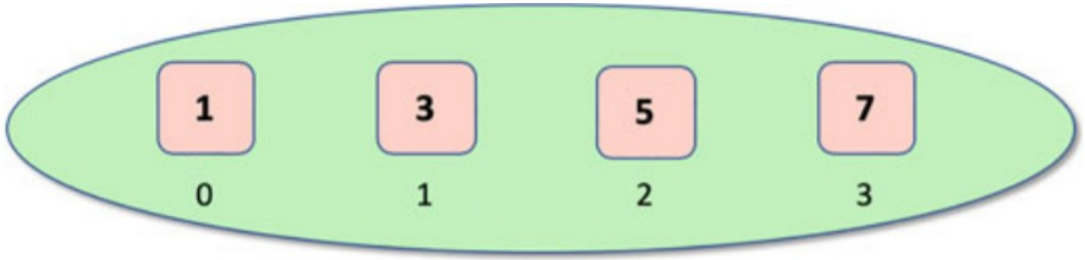
- Tuples
  - Tuple represents a ***collection of objects*** that are ***ordered and immutable***
  - Tuples ***allow duplicate*** members and are ***indexed***
- Lists
  - Lists hold a ***collection of objects*** that are ***ordered and mutable***
  - They are ***indexed*** and ***allow duplicate*** members
- Sets
  - Sets are a ***collection*** that is ***unordered and unindexed***
  - They are mutable but do not allow duplicate values to be held
- Dictionary
  - A dictionary is an ***unordered collection*** that is ***indexed by a key*** which ***references a value***
  - ***No duplicate keys*** are allowed
  - ***Duplicate values*** are ***allowed***
  - Dictionaries are mutable containers

# Tuples

[Hunt 2019]



- Tuples are an immutable ordered collection of objects
  - It is not possible to add or remove elements from the tuple once it has been created
- Tuples are defined using parentheses
  - *t = (1, 3, 5, 7)*
    - This tuple contains exactly *4* elements
    - The first element (the integer *1*) has the index *0* and the last element (the integer *7*) has the index *3*




- Tuples can also contain a mixture of different types
  - *t = (1, "Hello", True, (1, 3, 5))*

# Some Tuple Operations and Methods

[Hunt 2019]



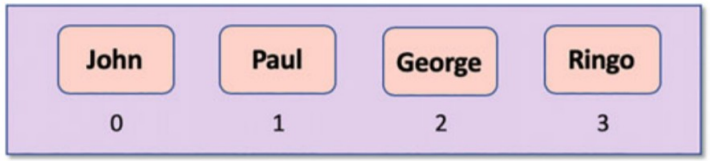
- Converting to tuple
  - *tuple([1, 2, 3])*
- Accessing elements of a tuple
  - *t[3]*
- Accessing a subset of elements of a tuple
  - *T[3:5]*
- Getting the length of a tuple
  - *len(t)*
- Counting an element in a tuple
  - *t.count("apple")*
- Finding the first index of a value in a tuple
  - *t.index("apple")*

# Lists

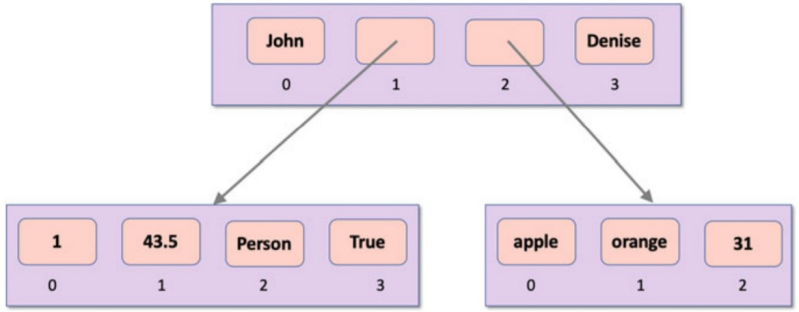
[Hunt 2019]



- Lists are ***mutable ordered containers*** of other objects
  - They support all the features of the tuple
  - Furthermore, they allow to add, remove, and modify elements
- Lists are created using square brackets
  - *l = [“John”, “Paul”, “George”, “Ringo”]*




- Similar to tuples, lists can contain a mixture of data types with a nested structure



# Some List Operations and Methods

[Hunt 2019]



- Converting to list
  - *list((1, 2, 3))*
- Adding elements to a list
  - *l.append(4)*
  - It changes the actual list
- Adding a list to a list
  - *l.extend([5, 6])*
  - *l += [5, 6]*
- Inserting elements into a list
  - *l.insert(2, 1000)*
- Removing elements by value from a list
  - *l.remove(4)*
- Removing elements by index from a list
  - *l.pop(2)*
- Reversing elements in a list
  - *l.reverse()*
- Sorting elements in a list
  - *l.sort()*

# Sets

[Hunt 2019]



- Set is an unordered (un indexed) collection of immutable objects that does not allow duplicates
- A Set is defined using curly brackets
  - *s = {1, 2, 3}*
- Sets can hold any immutable object
  - We cannot nest lists or other sets within a set as these are not immutable types

# Some Set Operations and Methods

[Hunt 2019]



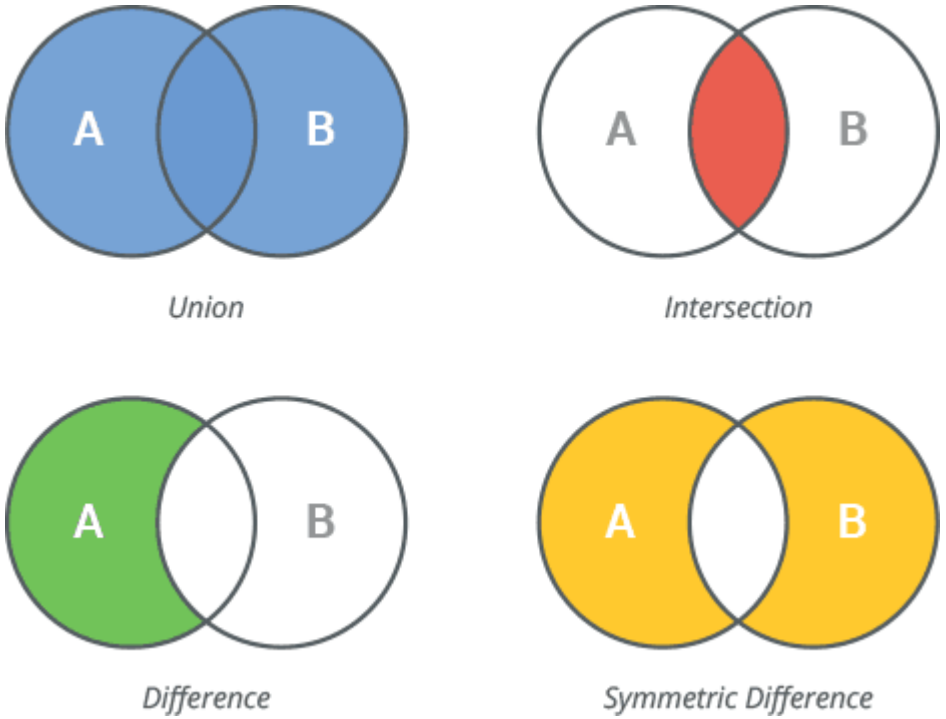
- Converting to set
  - *set([1, 1, 2, 3])*
- Checking for presence of an element
  - *print(1 in s)*
- Adding an item to a set
  - *s.add(4)*
- Adding multiple items to a set
  - *s.update([5, 6, 7])*
- Removing an Item from set
  - *s.remove(5)*

# Some Set Operations and Methods

[Hunt 2019]



- Union
  - *s1.union(s2)*
  - *s1 | s2*
- Intersection
  - *s1.intersection(s2)*
  - *s1 & s2*
- Difference
  - *s1.difference(s2)*
  - *s1 - s2*
- Symmetric difference
  - *s1.symmetric_difference(s2)*
  - *s1 ^ s2*

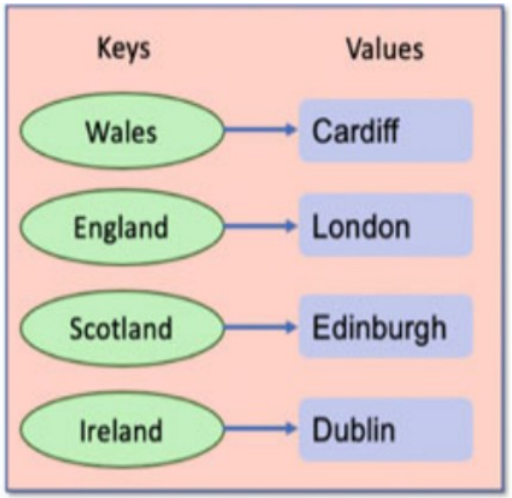


# Dictionaries

[Hunt 2019]



- A Dictionary is a set of associations between ***a key*** and ***a value***
  - The dictionary is unordered, changeable (mutable) and indexed
  - The keys must be unique, but the values do not need to be unique
  - Keys must be immutable
- A Dictionary is created using curly brackets ('{}') where each entry is a ***key:value*** pair
  - *d = {'Wales': 'Cardiff', 'England': 'London', 'Scotland': 'Edinburgh', 'Northern Ireland': 'Belfast', 'Ireland': 'Dublin'}*
  - The *dict()* function can be used to create a new dictionary object from an iterable or a sequence of *key:value* pairs
- A dictionary can be nested
  - The value itself can be a container



# Some Dictionary Operations and Methods

[Hunt 2019]



- Accessing items via keys
  - *d['Wales']*
- Adding a new entry
  - *d['France'] = 'Paris'*
- Changing a keys value
  - *d['Wales'] = 'Swansea'*
- Removing an entry
  - *d.pop("Wales")*
- Checking a key membership
  - *print("Wales" in d)*
- Accessing to keys, values, or key-value pairs
  - *d.keys()*
  - *d.values()*
  - *d.items()*

# Membership Operators

https://www.w3schools.com/python/python_operators.asp



- Membership operators are used to test if a sequence is presented in an object

| Operator | Description | Example |
|---|---|---|
| *in* | Returns ***true*** if a sequence with the specified value is present in the object | *x in y* |
| *not in* | Returns ***true*** if a sequence with the specified value is not present in the object | *x not in y* |

# Where to Use What?



- The time complexity of different operations on different collection types is different
  - For example, the time complexity of finding an element in a List is *O(n)* while it is *O(1)* in a dictionary
- Be careful to choose the right collection type
  - Check out “data structures and algorithms” textbooks for more information